

Digital Services Act

Transparency Report for Consumer Communication Products (Skype & Teams)

February 2025

Introduction

This DSA Transparency Report covers Skype and Teams. Skype and Teams are consumer communications products and are used by consumers for communications purposes, both to other Skype and Teams users and the mobile and landlines.

This DSA Transparency Report is responsive to the obligations under DSA Article 15(1), Article 24(1), and Article 42(2)-(3). This Report provides information regarding the following topics as they pertain to the European Union:

- Content Moderation at Microsoft's Initiative
- Content Moderation & Automated Systems
- Content Moderation following a User Report
- Content Moderation Appeals
- Government Requests
- Out of Court Settlement Body Disputes

Content Moderation at Microsoft's Initiative

Microsoft provides the information below in response to DSA Article 15(1)(c). This section reports data regarding content moderation Skype and Teams engaged in on their own initiative, absent a user report.

This section describes activities Skype and Teams undertake at its own initiative to address illegal content or information in violation of Skype's terms and conditions, which can be found Microsoft Services Agreement.

When Skype or Teams flag content through automated detection, the service may restrict the content owner's ability to edit or share that content while the content awaits human review.

Violating content is blocked from sharing or is removed, and, in the case of egregious violations such as sharing child sexual abuse material, the Microsoft account of the content owner may also be terminated in accordance with Microsoft's published policies.

Product Data, respectively: Teams | Skype

Part 1 of 2

Policy Violation	Number of measures taken at the provider's own initiative		Number of measures taken after detection with solely automated means	
Pornography or Sexualized Content	435		435	
Protection of Minors	44	7,769	44	7,769
Risk for public security	7	29	7	29
Violence	56		56	

Part 2 of 2

Policy Violation	Visibility restriction Removal		Account restriction Suspension	
Pornography or Sexualized Content	435			
Protection of Minors	27	5,531	17	2,238
Risk for public security	7	29		
Violence	56			

Content Moderation & Automated Systems

Skype and Teams provide the information below in response to DSA Articles 15(1)(e) and 42(2)(c).

Skype and Teams use automated content detection on images, videos and profile pictures when images or videos are shared by consumer accounts or when a profile picture is uploaded. Furthermore, Skype's automated detection system monitors profile first and last names to prevent impersonation, fraud, phishing, and scams.

Skype and Teams leverages automated content detection to detect known child sexual abuse material (CSAM) and terrorist content using both PhotoDNA and MD5 hashes.

PhotoDNA is a perceptual hashing technology, which is used to identify visually identical and near identical images and video. To do so, PhotoDNA creates a unique digital signature

(known as a “hash”) of an image or selected frames in a video and compares it against signatures (“hashes”) of previously identified CSAM and terrorist content images or video. PhotoDNA is not facial recognition software and cannot be used to identify a person or object in an image.

For CSAM, PhotoDNA is used to detect content that has previously been determined to sexualize a child based on a classification scale widely adopted across industry. The service draws from an internal database of Microsoft-created hashes for CSAM content, as well as hashes provided by trusted third-party organizations.

For terrorist content, PhotoDNA is used to detect content that has previously been determined to be violent extremist and terrorism content consistent with Microsoft policies and which has been hashed for inclusion in an internal database of Microsoft-created hashes, as well as hashes provided by the Global Internet Forum to Counter Terrorism (“GIFCT”).

Message-Digest Algorithm (“MD5”) is a broadly used cryptographic hashing technology, which is provided as part of standard programming libraries and is used to determine if two items of content are identical. Microsoft currently deploys MD5 in conjunction with PhotoDNA to identify CSAM and terrorist content.

A CVS content validation system is used to detect impersonation, fraud, phishing, and scam-related content. It works by comparing text strings against a predefined list of prohibited keywords commonly used by fraudsters to impersonate entities such as government officials.

Teams also uses automated text-scanning for ‘Community name and description’ (only) which prevents users from saving a Community name/description with inappropriate language. If the name/description is deemed inappropriate the user sees an error and is prompted to try something else right there (similarly to entering a wrong password when trying to log in); the content isn’t saved, and is thus prevented (vs. taken down).

For text-scanning, Teams uses SIFT (AKA TwoHat) which detects inappropriate language based on a proprietary service built by SIFT. The scanning results in risk-score across a spectrum of harms, and Teams has the ability to configure the risk-score threshold which results in preventing the user from saving the desired field.

The specifics of the technical implementation of PhotoDNA and MD5 hash matching are covered above.

The PhotoDNA distance threshold, which determines how “close” the Euclidian distance between two hashes needs to be in order to consider them a match, was selected based

on the largest distance for which there were no false positives as determined during PhotoDNA's initial validation, performed in partnership with the National Center for Missing and Exploited Children.

Given that MD5 is solely capable of identifying exact matches, there are no parameters set for its use.

Skype and Teams perform automated detection by generating PhotoDNA and MD5 hashes for the content to be scanned and comparing the resulting hash against the hashes of known bad content (as earlier described). If a match is found, the suspected CSAM or terrorist content is verified through human review, with the following exceptions:

- If additional content that is awaiting review has the same hash, only one such instance will be reviewed by a human reviewer and the result of that review will be applied to all content pending review with that same hash.
- If there have been at least ten earlier human reviews that confirmed CSAM content from a given content owner within the previous day, then all additional content from that same content owner that matches a CSAM hash will be automatically actioned on as confirmed CSAM without additional human review.

If content is confirmed to be violative as described above, Skype will block sharing or remove the content and may terminate the account in accordance with Microsoft's published policies. The service also notifies the content owner of the enforcement action and, if the content appears to be CSAM, reports it to the U.S. National Center for Missing and Exploited Children.

For Skype and Teams, CVS (Content Validation Service) is used to scan user content such as text on customer first and last name to detect and block harmful or offensive content. Here are some key points on how CVS works to block content:

Detection and Blocking: CVS uses two lists for text detection: a global list for scanning text content across all integrated product groups, and a custom list specific to each product group to meet their unique detection needs. When harmful content is detected, CVS blocks the content and sends notifications to relevant authorities.

Manual Moderation: In some cases, manual moderation is used to review and block content, especially when there is a risk of false positives. This approach helps in fine-tuning the solution and ensuring that legitimate content is not blocked unnecessarily.

Integration with Other Services: CVS is integrated with various Microsoft services, including Skype and Teams, to block specific keywords and harmful content. The service supports

both whole word and substring matching, although whole word matching is the standard configuration to avoid undesirable effects.

Notification and Compliance: Upon detecting a violation, CVS sends email notifications to users. For MSA (Microsoft Account) users, CVS signals the MSA team to take action against the account.

There are two primary contributors to the effectiveness of hash matching systems that leverage PhotoDNA or MD5: the correctness of the hashing algorithm itself (the technology) and the management of the set of hashes against which content is matched (the content). The technology itself is content agnostic.

The correctness of the technology is ensured by validating that any updates do not alter its results when tested against a known set of inputs (i.e., we confirm that the hash calculations return the same result).

Hashing algorithms leverage databases of reference hashes of previously identified CSAM and terrorist content images or videos when assessing potentially violative content in order to find copies of the same content. Given that PhotoDNA and MD5 hash values are guaranteed to be stable, the precision and recall of these algorithms is, by definition, 100% for a set of previously identified violative hashes.

The following information relates to how the performance of the content (i.e. reference hashes) is tested and monitored. This is separate and distinct from the performance of the technology and is impacted by match distances used and the quality and consistency of human reviews both by Microsoft and by the third parties that generate hashes that Microsoft leverages.

Microsoft's reference hashes are updated and maintained in the following ways:

- Human review improves the accuracy of the underlying hashes by considering what the reviewer determined.
- In addition, in the case of PhotoDNA, hashes that are very similar are grouped to ensure they provide the same result.

As an indicator of accuracy, we have reversed 158 instances of Terrorist and CSAM enforcement actions. There is no direct measure of recall, though it should be noted that the previously identified hashes represent a small but unknown fraction of all violative content.

As noted previously, human review is not leveraged in cases where it would have no impact on the outcome of the review process (such as the prevention of multiple reviews of duplicate content) or would have no significant impact on the resulting enforcement

actions (such as when the content owner's account has already been terminated for repeated confirmed violations). The measures of precision and recall of such actions would be equivalent to that of the underlying hash sets as described above.

Microsoft performs automated detection for child sexual abuse material (CSAM) because Microsoft is committed to protecting children from online harm. We do not allow the exploitation of, harm, or threat of harm to children on our services. This includes banning the use of our services to further child sexual exploitation and abuse (CSEA). When we become aware of content containing apparent child sexual exploitation and abuse, Microsoft reports the content to the National Center for Missing and Exploited Children (NCMEC).

CSEA is any content or activity that harms or threatens to harm a child through exploitation, trafficking, extortion, endangerment, or sexualization. This includes the sharing of visual media that contains sexual content that involves or sexualizes a child. CSEA also includes grooming, which is the inappropriate interaction with children by contacting, private messaging, or talking with a child to ask for or offer sex or sexual content, sharing content that is sexually suggestive, and planning to meet with a child for sexual encounters. A child is anyone under 18 years old.

Microsoft performs automated detection for terrorist content because we want to protect our users from illegal and harmful content and conduct. We also recognize that we have an important role to play in preventing terrorists and violent extremists from abusing online platforms. We do not allow content that praises or supports terrorists or violent extremists, helps them to recruit, or encourages or enables their activities. We look to the United Nations Security Council's Consolidated List to identify terrorists or terrorist groups. Violent extremists include people who embrace an ideology of violence or violent hatred towards another group.

In addressing terrorist and violent extremist content, we also work to ensure that people can use our services to talk about terrorism or violent extremism, share news or research about it, or express opposition to it.

Microsoft chooses not to perform human review on content with a hash that matches the hash of another content item that was already awaiting human review, because the matching hash value confirms that additional reviews of that same content provide no additional benefit.

Microsoft chooses not to perform human review of additional suspected CSAM content from a given content owner after at least ten content items from that owner have been confirmed by human review to be CSAM within the previous day because, based on

published policy, the associated account would have been terminated after the first such confirmation and, thus, the marginal value of reviewing additional violations is limited with respect to informing enforcement actions.

In both cases, the elimination of these low-value reviews reduces the frequency with which human reviewers are exposed to harmful content that could negatively impact their wellness.

Microsoft leverages a broader human review process in conjunction with hash-based automated content detection. A matched image is directed to a human moderation queue (depending on the harm type of the matching reference hash), where one or more human reviewers may independently review the content to confirm that the content is violative before enforcement action is taken.

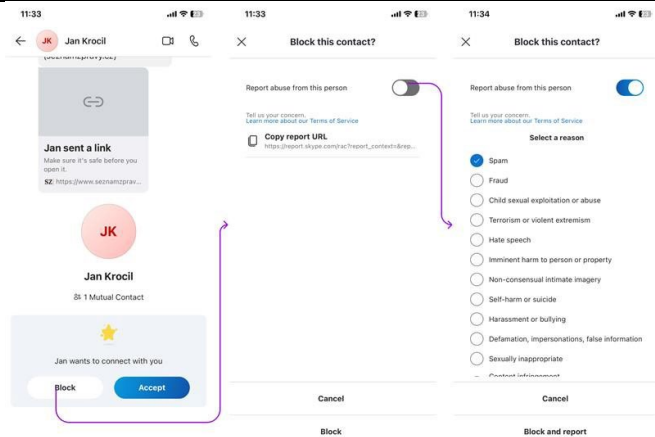
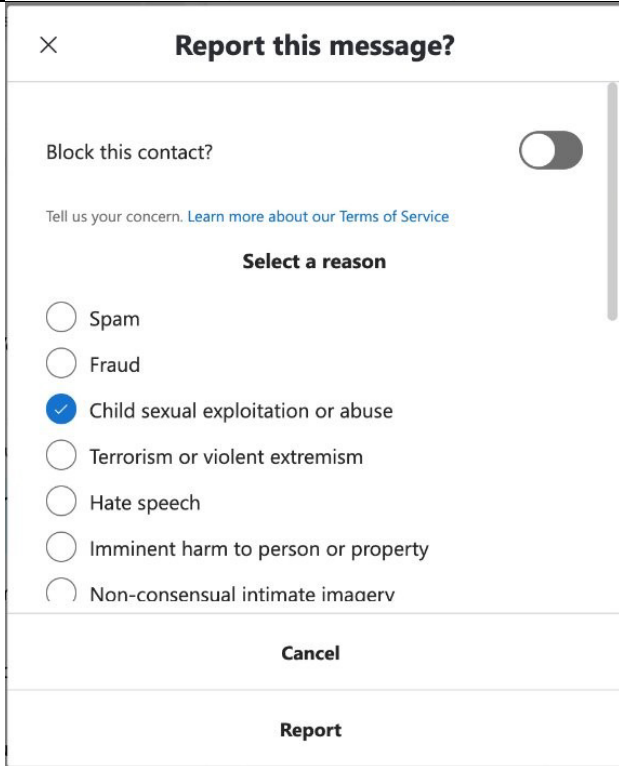
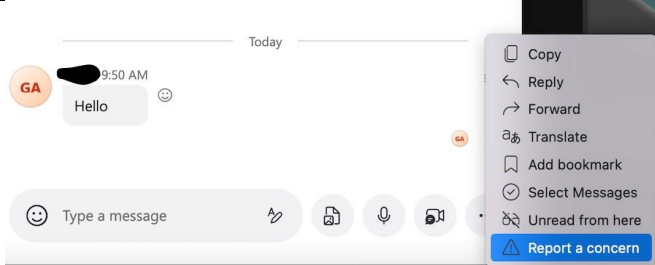
In cases where no human review occurs because content with the same hash was already in the process of being reviewed or the owning account had already been terminated as a result of at least ten other content items that had been reviewed by humans and confirmed to be CSAM, the impacted content owner can submit an appeal for the return of the content.

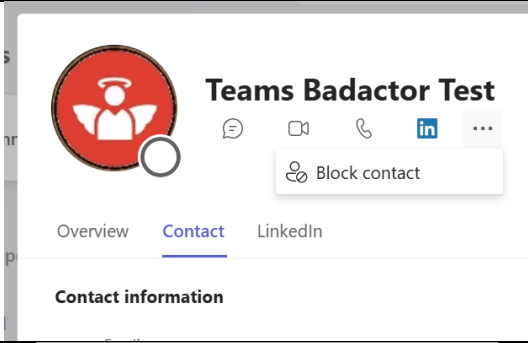
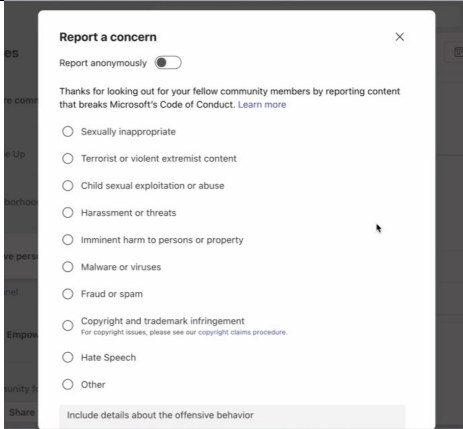
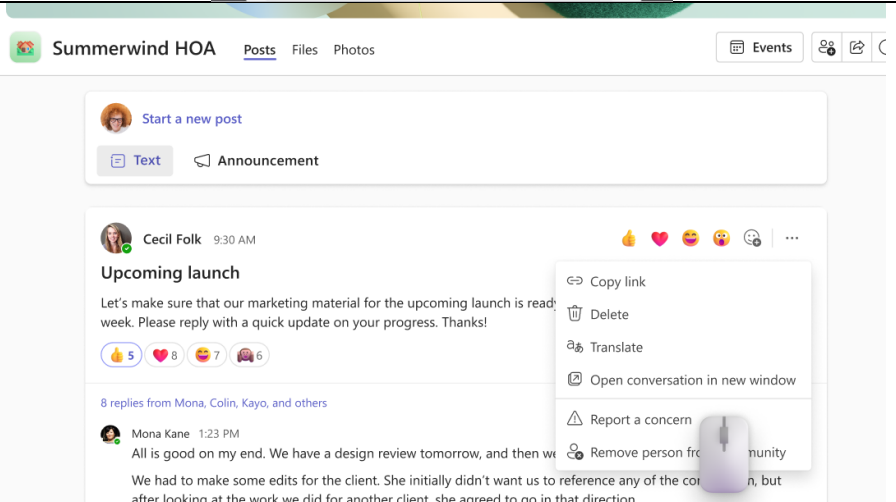
In the case of a user whose account is terminated following the detection of CSAM by automated means, the impacted user can submit an appeal to have the account returned. As an additional safeguard, Microsoft monitors the frequency with which such appeals result in the account suspension being reversed.

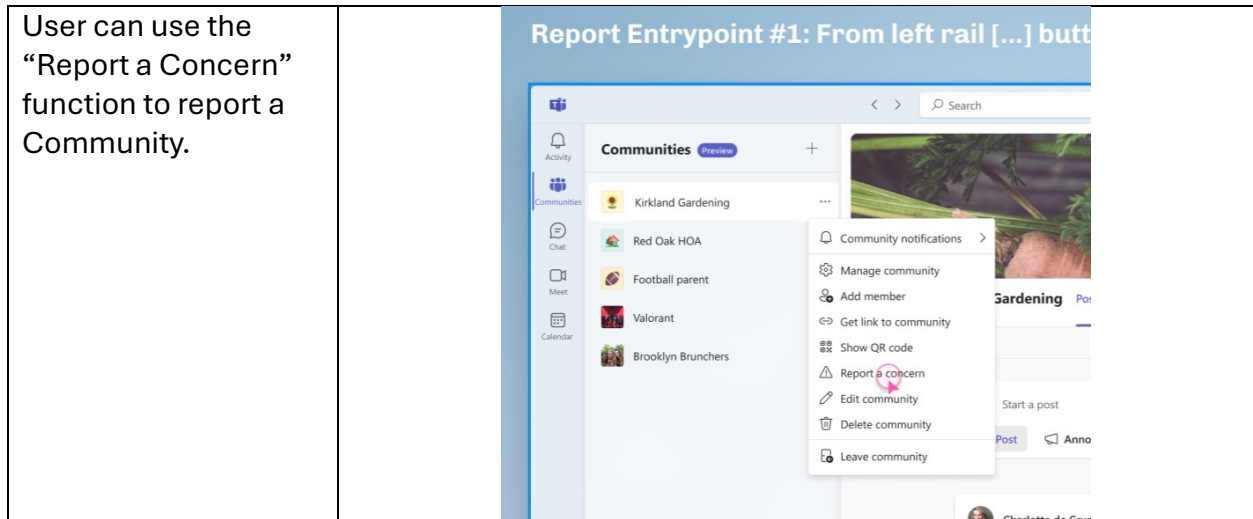
	Teams	Skype
Number of measures not taken by automated means	646	10,783
Number of measures solely taken by automated means	7	214

Content Moderation following a User Report

Microsoft provides the information below in response to DSA Articles 15(1)(b)-(c) and 41(2)(a)-(b).

SKYPE		
User has the ability to report another user and block.		
User has the ability to report a specific message.		
User can use the “Report a Concern” function to report content.		
TEAMS		

User has the ability to block another user.	
User has the ability to report a specific message.	
User can use the “Report a Concern” function to report content.	



Product Data, respectively: Teams | Skype

Part 1 of 2

Policy Violation	Number of notices received		Number of specific items of information included in the total number of notices	
Illegal or Harmful Speech	235	87,494	235	87,494
IP Infringements	65	1,370	65	1,370
Nonconsensual Behaviour	254	260,589	254	260,524
Pornography or Sexualized Content	47	8,748	47	8,748
Protection of Minors	131	140,045	131	140,045
Risk for public security	248	28,250	248	28,250
Scams and Fraud	1,446	638,594	1,446	638,594
Scope of Platform Services	1,777	359,147	1,777	359,145
Self Harm	81	57,853	81	57,853
Violence		2		2

Part 2 of 2

Policy Violation	Median time to take action		Number of actions taken on the basis of the terms and conditions of the service	
Illegal or Harmful Speech	68.2	12.7	9	

IP Infringements	86.3	68.5		
Nonconsensual Behaviour	2.8	357.2	5	2,903
Pornography or Sexualized Content	6.7	5.4	94	
Protection of Minors		56.9		219
Risk for public security		18.7		6
Scams and Fraud	83.0	99.6	1	71
Scope of Platform Services	4.3	178.2		
Self Harm	4.0	217.0	1	
Violence	1.0	31.5	1	

Content Moderation Appeals

Microsoft provides the information below in response to DSA Article 15(1)(d)

	Teams	Skype
Number of complaints submitted to the internal-complaints mechanism	15.0	2235.0

Government Requests

Microsoft provides the information below in response to DSA Article 15(1)(a)

Skype and Teams did not receive any applicable government requests during the reporting period.

Out of Court Settlement Body Disputes

Microsoft provides the information below in response to DSA Article 24(1)(a).

Skype and Teams for Life did not receive notice of any disputes submitted to an out-of-court dispute settlement bodies during the reporting period.